The compact E_8 adjoint sigma-MGF

Proof and dimension-aware verification strategy

Computational proof companion

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Abstract

We prove the exact constant-term formula for the 248-dimensional adjoint module, verify its root and zero-weight data and its quadratic invariant exactly, and prove the alternating cubic row algebraically. We also record the correct one-copy Chevalley series and explain why it does not determine the multivariate sigma-MGF.

1 Exact formula

For every unitary compact-group representation,

$$\mathcal{S}_{K,V} = \int_K \prod_a \det(I - t_a\rho(g))^{-1} dg = \sum_\lambda \dim(\mathbb{S}_\lambda V)^K s_\lambda(\mathbf{t}).$$

Equivalently, $[m_\tau]\mathcal{S}_{K,V} = \dim(\bigotimes_i \operatorname{Sym}^{\tau_i} V)^K$. The adjoint E_8 weights are the 240 roots and zero with multiplicity eight. Weyl integration therefore proves

$$\boxed{\mathcal{S}_{E_8,248} = \frac{1}{696729600} \operatorname{CT}_z \left[D_{E_8}(z) \prod_a (1-t_a)^{-8} \prod_{\alpha\in\Phi(E_8)} (1 - t_a z^\alpha)^{-1} \right].}$$

2 Universal adjoint tensors

For a compact simple Lie algebra, the Killing form κ is a unique symmetric quadratic invariant. In degree three,

$$c(x, y, z) = \kappa(x, [y, z])$$

is invariant, nonzero, and fully alternating. Moreover

$$(\bigwedge^3 \mathfrak{g})^G \cong \operatorname{Hom}_G(\bigwedge^2 \mathfrak{g}, \mathfrak{g}),$$

and for the simple Lie algebra $\mathfrak{e}_8$ the bracket spans this Hom-space. Chevalley restriction gives basic invariant degrees $2, 8, 12, 14, 18, 20, 24, 30$, so $(\operatorname{Sym}^3 \mathfrak{g})^G = 0$. Since the only invariant in $\mathfrak{g}^{\otimes 3}$ is alternating, the Schur type $(2, 1)$ also has multiplicity zero. Consequently

$$\boxed{\mathcal{S}_{E_8,248} = 1 + h_2 + e_3 + O_{\geq 4}.}$$

Chevalley restriction gives the one-copy series

$$\boxed{\mathcal{S}(t, 0, \ldots) = \prod_{d\in\{2,8,12,14,18,20,24,30\}} (1 - t^d)^{-1}.}$$

This specialization cannot see e_3 and therefore does not determine the full sigma-MGF. Thus the degree-at-most-three monomial signature is

$$([m_{(2)}], [m_{(1,1)}], [m_{(3)}], [m_{(2,1)}], [m_{(1,1,1)}]) = (1, 1, 0, 0, 1).$$

3 Executed quadratic check and optional sparse cubic confirmation

The notebook generates the E_8 root orbit (240 vectors), appends eight zero weights, and constructs a 248 by 248 diagonal unitary torus matrix. It checks the determinant integrand directly and integrates the quadratic Schur characters exactly with the Weyl-denominator identity.

λ	$\varnothing$	(1)	(2)	$(1,1)$
proposed	1	0	1	0
exact Weyl check	1	0	1	0

The computed dimension is 248, with 240 roots and eight zero weights; all quadratic entries pass. Floating-point work is confined to the illustrative torus matrix, while invariant multiplicities use rational weights and exact integer sums.

The cubic multiplicities $(\mathrm{Sym}^3, \mathbb{S}_{(2,1)}, \bigwedge^3) = (0,0,1)$ are therefore proved algebraically above. For an optional mechanical confirmation, construct only the zero-weight part of $\bigwedge^3 \mathfrak{e}_8$ and impose the eight simple-root raising operators. The expected nullspace is one-dimensional and represented by $c(x,y,z)$; the analogous expected nullities on Sym^3 and $\mathbb{S}_{(2,1)}$ are zero. This sparse route avoids materializing spaces of dimensions on the order of $\binom{248}{3}$. It is not reported as an executed computation in this artifact.

Reproducibility. Run `e8_verification.py` with marimo.